Supplementary Material

Attributing and Budgeting the Cost of Agentic AI Under Stated Uncertainty:
A Task-Level Total Cost of Ownership Model

Xiaochao Pu xcpu@umich.edu

This document supports the paper. S1 lists the case-study parameters, S2 the trace-processing rules, S3 the source register and the evidence behind each failure mechanism, and S4 the evaluation protocol history and the reproduction steps. Code and data are in the accompanying package (version 1.0.0).

S1. Case-study parameters and outputs

The simulation (Section VI of the paper) draws 200,000 samples with seed 20261004. All inputs are assumptions chosen for illustration. Table 1 gives every input, and the formulas follow.

Table 1: Inputs of the case-study simulation

Input	Distribution or value	Note
Steady-state volume	40,000 tickets/month $\times$ lognormal(0, 0.15)	One volume multiplier per draw
Ramp	Month 1: 1/3; month 2: 2/3; then 1	36-month horizon
Median tokens per ticket	140,000 $\times$ lognormal(0, 0.40)	Deployment-level uncertainty
Per-ticket spread	lognormal, $\sigma = \ln 3/1.645 = 0.668$	90% of tickets within a factor of 3 of the median; mean = median $\times e^{\sigma^2/2}$
Blended price	85% input at \$2.00, 15% output at \$12.00 per million tokens = \$3.50	October 2026 list prices
Price-capture multiplier m	Triangular(0.30, 0.70, 1.00)	Year 1: 1; year 2: m; year 3: m^2
Failure rate	Beta(6, 94), mean 6%	Share of tickets needing correction
Correction cost per failed ticket	Triangular(\$3, \$6, \$12)	Applied to every failed ticket
Drift migrations	$36/\text{cycle}$ events, cycle $\sim$ Uniform(12, 18) months	Cost per event Triangular(\$20k, \$35k, \$70k)
Strategy and selection (one-time)	Triangular(\$12.0k, \$18.4k, \$30.0k)	
Implementation (one-time)	Triangular(\$60k, \$86k, \$130k)	
Support (per year)	Triangular(\$18k, \$22k, \$35k)	Drawn separately each year
Measurement setup (one-time)	Triangular(\$25k, \$40k, \$70k)	
Measurement run (per year)	Triangular(\$15k, \$25k, \$45k)	Drawn separately each year

Formulas. With V_y the tickets in year y, $\bar{k}$ the mean tokens per ticket, p_0 the blended price and m_y the price multiplier ($m_1 = 1$, $m_2 = m$, $m_3 = m^2$): $\text{inference}_y = V_y\,\bar{k}\,p_0\,m_y/10^6$; $\text{rework}_y = V_y \cdot f \cdot c$ with failure rate f and correction cost c; drift $= (36/\text{cycle})\times$ cost per event. Three-year TCO is the sum of inference, rework, drift, support, measurement, strategy and implementation. Inputs are drawn independently.

Outputs. Three-year TCO: P5 \$1.06M, P50 \$1.60M, P95 \$2.53M; mean \$1.67M; P80 \$2.02M; P90 \$2.28M. A billing-only estimate (median tokens, flat list price, no rework, drift or measurement) is about \$0.87M. Per ticket, the median-token cost is \$0.49 and the mean \$0.61; the top 1% of tickets account for 4.9% of spend.

Unit cost. Accepted outcomes are tickets not reopened within 30 days (acceptance rate 94%). With shared cost allocated in proportion to accepted outcomes, median cost per accepted outcome is \$0.90 avoidable and \$1.22 fully loaded. A 7-day window with a 4% failure rate gives \$0.74 and \$1.05.

Table 2: One-at-a-time sensitivity of median three-year TCO

Input varied	Low case	High case
Volume (tickets/month)	20,000: $1.01M	80,000: $2.78M
Blended price	Low-tier ($0.10/$0.40): $0.98M	Flagship ($5/$25): $2.38M
Median tokens per ticket	70,000: $1.29M	280,000: $2.21M
Failure rate	3%: $1.31M	10%: $1.99M
Correction cost	Halved: $1.31M	Doubled: $2.15M
Price-capture multiplier	(0.5, 0.9, 1.0): $1.70M	(0.15, 0.4, 0.7): $1.44M

S2. Trace processing and cost rules

S2.1 Sources and scope

Per-task costs come from Holistic Agent Leaderboard (HAL) traces, downloaded on 11 October 2026. Table 3 lists the sets. File names, run IDs and SHA-256 hashes are in `data/PROVENANCE.md`. Each task has one run.

Table 3: Data sets used in the pilot

Benchmark	Agent scaffold	Model	Tasks	Groups
USACO	Episodic + Semantic	DeepSeek V3, GPT-4.1, Gemini 2.0 Flash, Claude 3.7 Sonnet	307	bronze 123, silver 100, gold 63, platinum 21
SWE-bench Verified Mini	SWE-Agent v1.0	DeepSeek V3, Gemini 2.0 Flash, Claude Opus 4.1	50	sphinx 25, django 25

S2.2 Counting rule

HAL logs some calls twice: an inner provider span (for example an OpenAI or Anthropic record) nested under an outer `litellm` span. Counting both doubles tokens and cost. We count only the outer `litellm` span of each call. When the span summary carries provider usage we use it, and otherwise we use the output usage. Tasks are identified by the `weave_task_id` attribute.

S2.3 Pricing rule

Cost is tokens times the dated list price in Table 4. For Anthropic models, fresh input tokens are prompt tokens minus cache-read tokens; cache-read tokens are priced at 0.1 times and cache-creation tokens at 1.25 times the input price, and cache-creation tokens are reported separately from prompt tokens in these files. Prices are assumptions dated to the 2025 runs and are not read from the traces.

Table 4: List prices applied (USD per million tokens)

Model	Input	Output
DeepSeek V3	1.25	1.25
GPT-4.1	2.00	8.00
Gemini 2.0 Flash	0.10	0.40
Claude 3.7 Sonnet	3.00	15.00
Claude Opus 4.1	15.00	75.00

S2.4 Validation against HAL totals and data-quality findings

Applied to HAL's reported (double-counted) usage, the prices reproduce HAL's reported total cost for the four USACO models and for DeepSeek V3 and Gemini 2.0 Flash on SWE-bench. Table 5 compares HAL's reported totals with totals recomputed from unique calls.

Table 5: USACO: reported versus recomputed totals

Model	HAL total (USD)	Recomputed (USD)	Token ratio	Accuracy
DeepSeek V3	12.08	6.04	2.0	0.391
GPT-4.1	28.10	14.05	2.0	0.450
Gemini 2.0 Flash	1.46	0.73	2.0	0.270
Claude 3.7 Sonnet	38.70	38.70	1.0	0.293

Three further findings affect interpretation. First, the SWE-bench Opus 4.1 file reports $131, and list prices applied to the recorded tokens give $187; the reported total omits 2.96 million cache-creation tokens. Second, every SWE-bench run used a $4 per-instance cost limit, and 64% of Opus 4.1 tasks cost at least $3.90, so those costs are censored. Third, HAL computes leaderboard cost without caching benefits, so the same run can carry different costs in different HAL views. HAL's USACO page carries a known-issue notice dated 5 October 2026 about the double counting.

S2.5 Per-task cost shape

The 90th-to-50th percentile cost ratio is 1.26 (Claude 3.7 Sonnet), 1.29 (Gemini 2.0 Flash), 1.66 (GPT-4.1) and 1.74 (DeepSeek V3) on USACO, where the agent makes about two model calls per task. On SWE-bench it is 1.05 (Opus 4.1, censored), 2.04 (DeepSeek V3) and 11.77 (Gemini 2.0 Flash). Rank correlation of per-task cost across models is 0.30–0.50 on USACO (307 tasks in common) and 0.17–0.28 on SWE-bench (50 tasks).

S3. Source register and evidence for the failure mechanisms

S3.1 Source register

Table 6 lists the sources behind the qualitative synthesis. "Type" distinguishes the kind of claim each source can support. The accessed column is to be completed for each source before submission; the dates of the documents themselves are as cited in the paper.

Table 6: Source register

Key	Source	Type	Used for	Accessed
aws2026	AWS Well-Architected, Agentic AI Lens (Jun. 10, 2026)	Provider documentation	M1–M6, M9; cost per task completion; tag taxonomy; trace IDs	[to fill]
finout2026	Finout, FinOps for AI agents (Apr. 27, 2026)	Vendor guidance	M1, M7	[to fill]
prefactor2026	Prefactor, agent cost attribution (Jun. 13, 2026)	Vendor guidance	Table I (platform practice)	[to fill]
credyt2026	Credyt, per-task attribution (Jul. 7, 2026)	Vendor-authored	Attribution as engineering work	[to fill]
ibm2026, apptio2026	IBM Newsroom (Aug. 6, 2026); Apptio case study	Vendor documentation	Table I (enterprise TCO suite); M8	[to fill]
focus2026	FinOps Foundation, FOCUS v1.4, v1.5 plan	Standards body	Table I; token-field alignment (v1.5 unratified)	[to fill]

Key	Source	Type	Used for	Accessed
tbm2026	TBM Council, Taxonomy 5.0 and TBM for AI	Standards body	Table I	[to fill]
finopsue	FinOps Foundation, unit economics	Standards body	Unit economics practice	[to fill]
gartner2025	Gartner press release (Jun. 25, 2025)	Analyst forecast	Introduction (project cancellation forecast)	[to fill]
angelo2026	Fortune (May 26, 2026)	Press	Introduction; M4, M8	[to fill]
openai2026	OpenAI API deprecations policy	Provider documentation	Drift timing	[to fill]
epoch2025	Epoch AI data insight	Empirical	Price decline, 9×–900× per year	[to fill]
gundlach2025	arXiv:2511.23455	Empirical	Price decline, 5–10× per year	[to fill]
bai2026	arXiv:2604.22750	Empirical	Token variance; self-prediction; accuracy at intermediate spend	[to fill]
casey2026	arXiv:2604.27082	Empirical	Migration cost; model deprecation cadence	[to fill]
kapoor2025	arXiv:2510.11977	Empirical	HAL traces	11 Oct 2026 (traces)
gao2020, aace118	GAO-20-195G; AACE RP 118R-21	Costing guidance	Budget at a stated confidence level; limits of estimate ranging	[to fill]
martens2012	Hawaii Int. Conf. System Sciences, 2012	Peer-reviewed	Base TCO model	—

The Bai et al. code repository (https://github.com/LongjuBai/agent_token_consumption_analysis) was checked on 13 October 2026. It contains analysis scripts and prompt templates and does not contain the trajectories or per-task cost data.

S3.2 Evidence behind each mechanism

Each mechanism is listed only when at least one source states it. Table 7 gives the statement relied on and the type of evidence. Provider and vendor statements are descriptions of practice and do not measure how often a problem occurs.

Table 7: Mechanism, statement relied on, and evidence type

	Mechanism	Statement relied on (source)	Evidence type
M1	Billing grain differs from spend grain	Costs span reasoning cycles, multi-agent workflows and tool calls that standard billing cannot capture (aws2026); consolidated invoice lines are a structural gap (finout2026)	Provider and vendor documentation
M2	Tagging discipline degrades	Tag taxonomies exist on paper but are applied inconsistently, which breaks per-agent attribution (aws2026)	Provider documentation
M3	No trace propagation across agents	Without a workflow trace ID, workflows are analyzed one agent at a time and orchestration cost is inflated unnoticed (aws2026)	Provider documentation
M4	Cost and value recorded separately; no baseline	ROI practice requires the manual-process baseline (aws2026); usage not linked to useful features (angelo2026)	Provider documentation; press report

	Mechanism	Statement relied on (source)	Evidence type
M5	Detection slow and generic	Overruns found through monthly bills and heuristic 2σ/3σ thresholds without cited empirical basis (aws2026)	Provider documentation
M6	Optimization needs quality data	A cheaper workflow that degrades quality may cost more in business terms (aws2026); accuracy peaks at intermediate spend (bai2026)	Provider documentation; empirical study
M7	Prices and allocation rules become outdated	Monthly model retirement cadence (finout2026); about 12-month proprietary-model deprecation (casey2026); price decline of $9\times$–$900\times$ per year (epoch2025)	Vendor guidance; empirical studies
M8	Incentives and decentralized adoption	Internal usage leaderboard preceded budget exhaustion (angelo2026); five teams asked to build the same chatbot (apptio2026)	Press report; vendor case study; partly our synthesis
M9	Solutions are vendor-specific	AWS guidance is expressed in AWS services and did not mention FOCUS in the text searched; platforms use proprietary allocation logic (aws2026, finout2026)	Provider and vendor documentation; partly our synthesis

S4. Evaluation protocol history and reproduction

S4.1 Protocol history

Table 8 records what was specified when. The protocol and amendments were fixed before any real-data run. The recalibration test was added after the first results.

Table 8: Status of each evaluation component

Component	Status	Reading
E1, same-mix coverage (four budgets, $n \in \{10, 100\}$, 80% and 90% levels)	Pre-specified; amended 11 Oct 2026 before any real-data run	Primary evidence for RQ3
E1, shifted-mix coverage	Pre-specified; no threshold	Sensitivity to a change in task mix
E1c, recalibration after a shift	Post hoc	Exploratory
Repeat-run variability	Not executed (no repeat-run data)	Unresolved
Reference-class variant	Not executed	No evidence
E1b, price-path band test	Not executed	No evidence
E2, mapping completeness; E3, practitioner review	Not executed	Planned

The amendments of 11 October 2026 were the addition of the 80th percentile, the addition of the reference-class variant and the removal of $n = 1000$. The pre-specified criterion is coverage within ± 3 points of nominal on the unshifted split for the two-level bootstrap budget.

S4.2 Budget methods

For a batch of n tasks, with calibration task costs c_i, mean $\bar{c}$ and standard deviation s: (a) mean budget $n\bar{c}$; (b) normal budget $n\bar{c} + z_q s\sqrt{n}$; (c) resampled-quantile budget: draw $B = 1000$ batches of n tasks with replacement from the calibration tasks (one random run per task), and take the order statistic of rank $\lceil (B+1)q \rceil$ of the batch totals; (d) two-level budget: as (c), but each batch first resamples the calibration tasks with replacement and then draws n tasks from that resample.

Coverage is the share of 1000 resampled test batches whose total does not exceed the budget. Each setting averages 20 random splits (unshifted: 153 calibration and 154 test tasks of 307). The shifted split calibrates on the cheaper half of the groups ranked by mean cost and tests on the costlier half. The per-task cap is the split-conformal quantile of rank $\lceil (m+1)q \rceil$ of m calibration costs.

S4.3 Reproduction

1. Install Python 3.10 or later with numpy 2.2.6 and pandas 2.3.3.
2. Run `bash run_all.sh`. It writes results to `out/`, takes under a minute, and exits with an error if any regenerated result differs from the released file.
3. Read `out/summary.txt` for every number quoted in the pilot section and `out/casestudy_output.txt` for the case-study sensitivity values.
4. To rebuild the per-task tables from the original traces, follow the README: download the files listed in the provenance note, decrypt them with HAL's published procedure, and run the extraction scripts.

The seed is fixed at 20261004 in every script. The HAL traces are not redistributed.